# MPU6050

6-axis motion sensor: accelerometer + gyroscope

The MPU6050 is a small motion sensor that combines a **3-axis accelerometer** and a **3-axis gyroscope** on one chip. It tells a microcontroller how the sensor is **tilted**, how fast it is **rotating** and how much it is **shaking**. It talks over I2C.

| | |
|---|---|
| **Maker** | InvenSense (now part of TDK) |
| **Sensors** | 3-axis accelerometer, 3-axis gyroscope, temperature sensor |
| **Accelerometer** | Ranges +/-2, 4, 8 or 16 g (16384 to 2048 counts per g) |
| **Gyroscope** | Ranges +/-250, 500, 1000 or 2000 °/s (131 to 16.4 counts per °/s) |
| **Converter** | Six 16-bit ADCs |
| **Interface** | I2C, address 0x68 (AD0 low) or 0x69 (AD0 high). Up to 400 kHz. |
| **Extras** | On-chip Digital Motion Processor (DMP), 1024-byte FIFO, INT pin, auxiliary I2C (XDA/XCL) for a magnetometer |
| **Power** | Chip 2.375 to 3.46 V. The GY-521 breakout has a regulator. We use 3.3 V. |

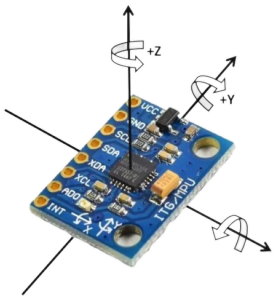



Fig 1. Axis directions on the module.

## How it works

- **Accelerometer.** A tiny mass on silicon springs moves when the chip accelerates. The chip measures that movement on X, Y and Z. Gravity is always pulling, so when the sensor is still, the readings show **which way is down**. That gives tilt (Fig 3).
- **Gyroscope.** Tiny vibrating masses feel a sideways (Coriolis) force when the chip rotates. The chip reports **rotation speed** in degrees per second. Adding it up over time gives an angle, but small errors build up (**drift**).
- **Together.** The accelerometer is steady over time but noisy during movement. The gyroscope is smooth in the short term but drifts. Mixing both (a complementary or Kalman filter, or the on-chip DMP) gives a stable orientation.
- **Reading it.** Convert counts to real units: m/s$^2$ = counts / sensitivity x 9.81. At rest, the three axes combine to about 9.8 m/s$^2$.

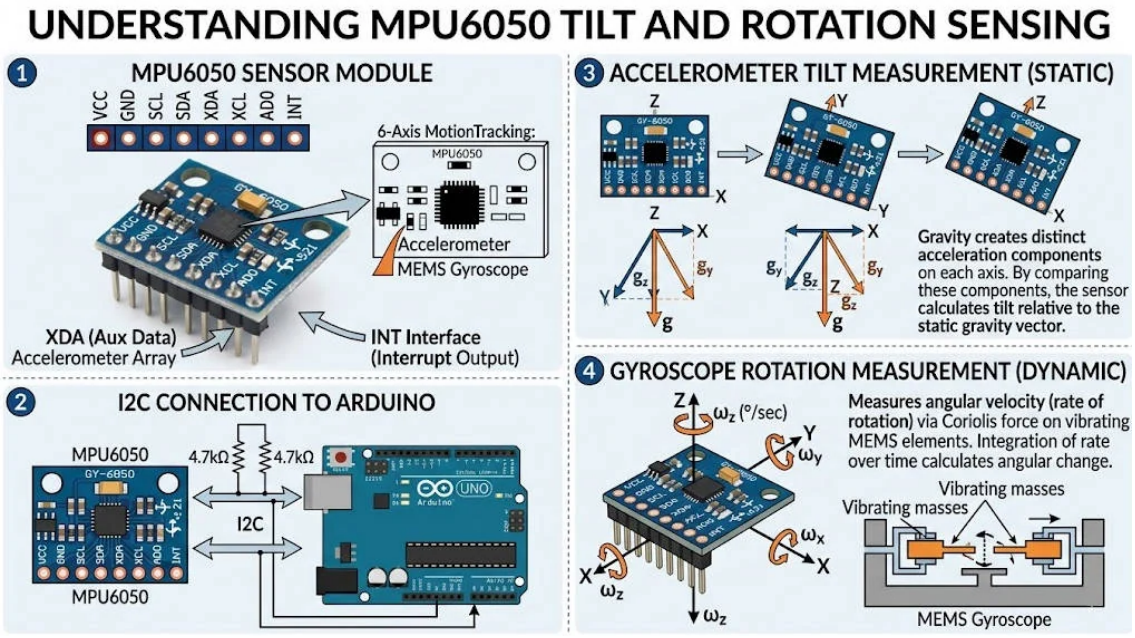



Fig 2. Overview: sensor pins, I2C hookup, tilt from gravity, and rotation measurement.

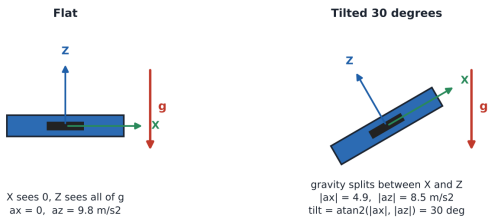



Fig 3. Tilt from gravity: the same 9.8 m/s$^2$ splits differently between the axes as the board tilts.

## Inside the chip, in one line

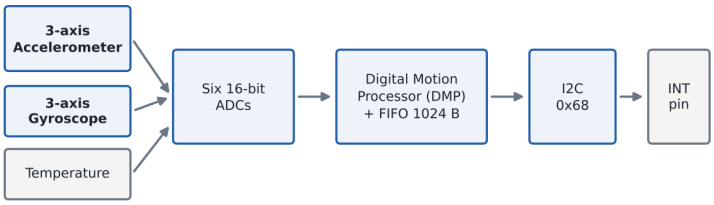



Fig 4. Signal path from sensors to your microcontroller.

## Pins and wiring

| Pin | What it does | Connect to (ESP32 example) |
|---|---|---|
| **VCC** | Power input. Use 3.3 V (the GY-521 board has a regulator). | 3V3 |
| **GND** | Ground | GND |
| **SDA** | I2C data line | GPIO21 |
| **SCL** | I2C clock line | GPIO22 |
| **AD0** | Sets the I2C address: low = 0x68, high = 0x69. Leave open or low for 0x68. | Leave unconnected |
| **INT** | Optional interrupt (data ready, motion). | Leave unconnected |
| **XDA, XCL** | Auxiliary I2C bus, to attach a magnetometer. | Leave unconnected |

Only four wires are needed: VCC, GND, SDA and SCL.

## Where it is used

| Used today | Could be used for |
|---|---|
| • Phones and tablets (screen rotation, games)<br>• Drones and RC aircraft (flight stabilisation)<br>• Self-balancing robots<br>• Game controllers and VR head tracking<br>• Camera gimbals and stabilisers<br>• Wearables (step counting, activity) | • Driver head-nod, head-tilt and distraction detection<br>• Hand tremor and steadiness tests<br>• Vibration and condition monitoring of motors, fans, pumps and conveyors<br>• Man-down and fall detection for lone workers<br>• Tilt, rollover and impact logging on vehicles and machinery<br>• Posture and physiotherapy tracking<br>• Tilt of floating sensor buoys and slopes |

## Know the limits

- **Movement fools tilt.** A braking or turning car adds acceleration, so the tilt reading is only trustworthy when things are fairly steady. Test parked first. Fix the board firmly, since a loose board adds false motion.
- **Gyro drift.** Angles from the gyro creep over time. Heading (yaw) cannot be corrected without a magnetometer.
- **Range choice.** Use +/-2 g or +/-4 g for tilt and tremor. Strong vibration needs +/-8 g or +/-16 g, or the readings clip.

Sources: MPU6050 datasheet explained (Ultra Librarian), summarising the InvenSense MPU-6000/6050 datasheet. Figures 3 and 4 are original diagrams.